

Rock Jobs: Which Property Matters?

BUILD • Science • 40 minutes

I can identify which tested or observed rock property matters for a stated everyday job.

Name: _____ Date: _____

You may write, type, say, sign, point or direct a partner. An adult can read the words and record your exact response. Keep the choice and explanation your own.

Words for this lesson

Word	Meaning
property	what the material is like
purpose	the job it must do
suitable	works for this job

First idea

A pupil says, 'The least permeable sample must be best for every rock job.' Which first idea would you test?

- A. Agree. Every use needs the least water passage.
- B. Ask what the job is; a rain barrier and a drainage layer need different water behaviour.
- C. Ignore water and choose the brightest colour for every job.

My first idea and reason:



A roof job card

We do • choose, explain, check

Which property question is most relevant to a rain-facing roof-cover model?

A. Which sample is the darkest?

B. Which prepared sample lets less water through and can form overlapping flat pieces?

C. Which sample has the longest name?

My choice: _____ Evidence or reason:

Job-to-property question lab

Route each brief clue to WATER CONTROL, WEAR / SURFACE, SHAPE / APPEARANCE or NEEDS ANOTHER CHECK.

Read or hear one card. Choose a group. Give the evidence. A partner checks your reason before you move on. Point or write card numbers; cutting is optional.

Group	Card numbers and evidence
WATER CONTROL	
WEAR / SURFACE	
SHAPE / APPEARANCE	
NEEDS ANOTHER CHECK	

Activity cards

1 • Rain-cover clue	2 • Drainage clue
The material must help limit water reaching the model space below.	Water should move away through spaces in a bed of suitable rock pieces instead of pooling at the top.
3 • Busy-path clue	4 • Walking-surface clue
The surface faces repeated shoes and wheels in the fictional yard.	The brief needs a stable, even surface; a separate local safety check would assess grip and installation.

5 · Overlap clue

The roof model needs thin pieces that can lie flat and overlap.

6 · Display clue

The brief is for a stable indoor display where pattern and finish are part of the stated purpose.

7 · Dark-colour claim

The darkest rock must be strongest, hardest and least permeable.

8 · Hardest-is-best claim

The sample with the faintest scratch mark should be used for every job, even without water or shape evidence.

Say: The job needs ___. The property question is ___ because ___. We still need to check ___.

Your lesson record

Create property-question cards for four everyday rock jobs before any candidate sample is chosen.

Model helps us see

Use the chain JOB → PROPERTY QUESTION → EVIDENCE NEEDED. Do not choose a sample until all three boxes are clear.

Model does not show

The briefs simplify real design. Cost, thickness, treatment, strength, installation, safety and local standards also matter; classroom evidence cannot approve real construction.

Supported

Match one key property picture to three clear jobs and complete one because sentence.

Use: For a __, we need to know __ because __.

Standard

Complete all four briefs, naming the relevant property evidence and one missing check for each.

Keep separate columns for job, property question and evidence needed.

Stretch

Explain a two-property trade-off and why opposite permeability behaviour can suit different water jobs.

Include one real-design factor the classroom brief does not model.

Evidence target

Four job-property questions, two because explanations, one needs-another-check decision and one model limit.

Exit, Audience and evidence • W12A

Supported: Complete or point: For a __, the property that matters is __.

Standard: Explain which property matters for one job and why another property does not answer it.

Stretch: Compare a rain-barrier and drainage purpose, explaining why useful permeability can point in opposite directions.

What I said, and what it changed

Next I will _____.

Adult witness note

What was observed, and with what level of independence?

My evidence and explanation

What does this evidence not show?

Visual reference cards

These are diagrams or model views from the uploaded lesson. Use their labels and the card descriptions; they are not photographs of your own results.



A path job card



A drainage job card

Exit • one next step

After the adult has listened, what will you keep, improve or check next?
